

Two operator-trace ratios from V_{600} K-class structure: 13/12 and 11/12, and their match to the H_0 and S_8 benchmarks

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Status: Preprint (not peer-reviewed). This paper has a deliberate three-layer structure: Layer 1 is exact mathematics, Layer 2 is a numerical observation from published data, Layer 3 is an explicit hypothesis. We do NOT claim Layer 2 or Layer 3 are theorems.

Abstract

We define two operators on the 12-dimensional rational T_τ -cycle space $C_{\mathbb{Q}} \cong \mathbb{Q}^{12}$ of the 600-cell: $\hat{H} := I_{C_{\mathbb{Q}}} + P_{72}$ and $\hat{C} := I_{C_{\mathbb{Q}}} - P_0$, where P_K is the diagonal rank- m_K projector onto the K-class cycles in the T_τ -cycle decomposition with K-multiset $\{72:1, 0:1, 52:5, 20:5\}$ (Paper 1, [Smart, 2026a](#)). We prove (Theorem 4.1) the exact rational identities $\text{tr}(\hat{H}) = 13$ and $\text{tr}(\hat{C}) = 11$, and we prove (Theorem 4.2) that within the K-saturated diagonal projector algebra of $C_{\mathbb{Q}}$ the only rank-one projector corrections to $I_{C_{\mathbb{Q}}}$ are $\pm P_{72}$ and $\pm P_0$. Both theorems are phase-independent across the $\mathbb{Z}_2^5 = 32$ canonical lifts of τ_σ (Paper 3, [Smart, 2026b](#)).

As a numerical observation, we then report: $\frac{13}{12} \cdot H_0^{\text{Planck}} \approx 72.97 \text{ km s}^{-1} \text{ Mpc}^{-1}$ versus SH0ES 73.04 ± 1.04 [[Riess et al., 2022](#)] ($\approx 0.06\sigma$ residual), and $\frac{11}{12} \cdot S_8^{\text{Planck}} \approx 0.763$ versus KiDS-1000 $0.766^{+0.020}_{-0.014}$ [[Heymans et al., 2021](#)] ($\approx 0.24\sigma$ directional residual using the lower KiDS asymmetric error; we use the lensing parameter $S_8 = \sigma_8 \sqrt{\Omega_m/0.3}$ since published weak-lensing intervals are reported as S_8). We do NOT derive cosmology from operator traces; the cycle-mode coupling rule that would upgrade the numerical match to a derivation is stated in §6 as an explicit hypothesis. The trace identities are exact given the imported Paper 1 K-multiset and the definitions above.

1 Introduction

The Hubble tension — the $\sim 5\sigma$ disagreement between early-Universe ($H_0 \approx 67.4 \text{ km/s/Mpc}$ from Planck 2018 [[Planck Collaboration, 2020](#)]) and late-Universe ($H_0 = 73.04 \pm 1.04 \text{ km/s/Mpc}$ from SH0ES 2022 [[Riess et al., 2022](#)]) inferences of the present-day Hubble constant — and the related σ_8/S_8 tension between Planck and weak-lensing surveys (KiDS-1000 [[Heymans et al., 2021](#)], DES-Y3 [[DES Collaboration, 2022](#)]) have driven a large literature of proposed resolutions [[Di Valentino et al., 2021](#)].

This paper does *not* propose a cosmological model. Instead, we report two finite-group facts about the binary icosahedral group $2I = V_{600}$, and observe that the corresponding rational ratios numerically match published tension ratios. We organise the paper around three explicit layers of claim:

- **Layer 1 (Theorem-grade):** exact rational trace identities on the 12-dimensional rational T_τ -cycle space.

- **Layer 2 (Observation):** numerical match between those ratios and currently-published tension benchmarks.
- **Layer 3 (Hypothesis):** a coupling rule from the operators to Λ CDM observables; *stated as a hypothesis, not claimed to be derived.*

The reader is invited to evaluate Layers 1 and 2 on their own merits and to treat Layer 3 with appropriate scepticism.

2 The 12-dimensional cycle space and K-class projectors

We import the following from Paper 1 [Smart, 2026a]; the present paper does not re-prove any of these statements.

- $V_{600} = 2I$, the binary icosahedral group of order 120, viewed as the vertex set of the regular 600-cell.
- A choice of $\tau \in 2I$ with $\text{ord}(\tau) = 10$ gives the cycle decomposition $T_\tau : g \mapsto \tau g$ of V_{600} into 12 cycles each of length 10.
- Each cycle is assigned a non-negative rational K-class via the K-multiset $\{72:1, 0:1, 52:5, 20:5\}$. Concretely, of the 12 cycles: one has K-class 72, one has K-class 0, five have K-class 52, and five have K-class 20.
- $H = \text{Dic}_5 \subset 2I$ is the bulk subgroup formed by the $K = 72$ and $K = 0$ cycles ($|H| = 20$).

Definition 2.1 (Cycle space). *Let $C_{\mathbb{Q}}$ be the 12-dimensional rational vector space with basis $\{e_i\}_{i=1}^{12}$ indexed by the 12 T_τ -cycles. Equivalently $C_{\mathbb{Q}} \cong \mathbb{Q}^{12}$. The basis is determined by the deterministic cycle ordering supplied by `vfd_v600.group.build_state()`.*

Definition 2.2 (K-class projectors). *For each $K \in \{72, 0, 52, 20\}$, the K-class projector is the diagonal operator $P_K \in \text{End}_{\mathbb{Q}}(C_{\mathbb{Q}})$ defined by*

$$P_K(e_i) = \begin{cases} e_i & \text{if cycle } i \text{ has K-class } K, \\ 0 & \text{otherwise.} \end{cases}$$

The identity operator on $C_{\mathbb{Q}}$ is denoted $I_{C_{\mathbb{Q}}}$.

The ranks of the four K-class projectors are exactly the K-multiset multiplicities:

$$\text{rk } P_{72} = 1, \quad \text{rk } P_0 = 1, \quad \text{rk } P_{52} = 5, \quad \text{rk } P_{20} = 5,$$

and they sum to $I_{C_{\mathbb{Q}}}$: $P_{72} + P_0 + P_{52} + P_{20} = I_{C_{\mathbb{Q}}}$.

Remark 2.1 (Baseline rule). *We define the Λ CDM-baseline operator as $I_{C_{\mathbb{Q}}}$ itself: the **K-blind uniform cycle weighting** that assigns coefficient 1 to every cycle basis vector. Once the baseline is so defined, $I_{C_{\mathbb{Q}}}$ is unique by construction. We do not claim $I_{C_{\mathbb{Q}}}$ is the unique τ_σ -invariant rank-12 operator on $C_{\mathbb{Q}}$ — many other operators commute with the τ_σ -induced cycle action; the baseline is a definitional choice, not a uniqueness theorem.*

3 The two operators $\hat{H}$ and $\hat{C}$

Definition 3.1 (Operators).

$$\hat{H} := I_{C_{\mathbb{Q}}} + P_{72}, \quad \hat{C} := I_{C_{\mathbb{Q}}} - P_0.$$

$\hat{H}$ is the max-K-anchor enhancement of the baseline ($\hat{H}(e_i) = 2e_i$ on the unique K-class-72 cycle and e_i otherwise). $\hat{C}$ is the trivial-K-class suppression of the baseline ($\hat{C}(e_i) = 0$ on the unique K-class-0 cycle and e_i otherwise).

The two operators are diagonal in the cycle basis with entries in $\{0, 1, 2\}$. Both are self-adjoint with respect to the standard $\mathbb{Q}$ -bilinear form on $C_{\mathbb{Q}}$.

4 Trace theorem (Layer 1)

Theorem 4.1 (Exact trace identities).

$$\text{tr}(\hat{H}) = 13, \quad \text{tr}(\hat{C}) = 11.$$

Equivalently, $\text{tr}(\hat{H})/12 = 13/12$ and $\text{tr}(\hat{C})/12 = 11/12$, both exact rationals.

Proof. For any diagonal projector P_K , the trace equals its rank, which equals the K-class multiplicity m_K . The K-multiset gives $m_{72} = 1$ and $m_0 = 1$; trace is additive on diagonal operators, so

$$\text{tr}(\hat{H}) = \text{tr}(I_{C_{\mathbb{Q}}}) + \text{tr}(P_{72}) = 12 + 1 = 13,$$

$$\text{tr}(\hat{C}) = \text{tr}(I_{C_{\mathbb{Q}}}) - \text{tr}(P_0) = 12 - 1 = 11. \quad \square$$

Theorem 4.2 (K-saturated admissibility). *Let $\mathcal{A}_K \subset \text{End}_{\mathbb{Q}}(C_{\mathbb{Q}})$ denote the K-saturated diagonal projector algebra, namely the $\mathbb{Q}$ -subalgebra generated by $\{I_{C_{\mathbb{Q}}}, P_{72}, P_0, P_{52}, P_{20}\}$. Within $\mathcal{A}_K$, the only rank-one diagonal projector corrections to $I_{C_{\mathbb{Q}}}$ — that is, the only signed corrections of the form $I_{C_{\mathbb{Q}}} \pm Q$ with $Q \in \mathcal{A}_K$ a rank-one projector — are $I_{C_{\mathbb{Q}}} \pm P_{72}$ and $I_{C_{\mathbb{Q}}} \pm P_0$.*

Proof. The four projectors $\{P_{72}, P_0, P_{52}, P_{20}\}$ are mutually orthogonal idempotents that resolve the identity, so any $Q \in \mathcal{A}_K$ has the unique decomposition

$$Q = a_{72}P_{72} + a_0P_0 + a_{52}P_{52} + a_{20}P_{20}, \quad a_K \in \mathbb{Q}.$$

Idempotence $Q^2 = Q$ on the orthogonal decomposition forces $a_K^2 = a_K$ for each K , hence $a_K \in \{0, 1\}$. The rank of such a Q is the K-multiplicity sum $\sum_K a_K m_K$ over those K with $a_K = 1$. With the K-multiset $\{72:1, 0:1, 52:5, 20:5\}$, the only 0/1 coefficient choices giving rank exactly 1 are $(a_{72}, a_0, a_{52}, a_{20}) = (1, 0, 0, 0)$ and $(0, 1, 0, 0)$, yielding $Q = P_{72}$ and $Q = P_0$. The signed rank-one corrections to $I_{C_{\mathbb{Q}}}$ are therefore exactly $\pm P_{72}$ and $\pm P_0$. $\square$

Remark 4.1 (Hypotheses are essential). *Theorem 4.2 requires both stated hypotheses: “diagonal in the cycle basis” and “K-saturated”. Dropping either one introduces a large family of rank-one projectors outside $\mathcal{A}_K$ (for instance any rank-one orthogonal projector on $C_{\mathbb{Q}}$ that is not K-saturated, or the rank-one τ_{σ} -invariant projector onto $\text{span}_{\mathbb{Q}}(e_{52} + e_{20})$ for any τ_{σ} -paired pair of $K = 52, K = 20$ basis vectors — this projector commutes with the τ_{σ} cycle action but is not diagonal in the cycle basis and is not in $\mathcal{A}_K$). The narrow theorem above is the strongest form we claim.*

4.1 Phase-independence across the $\mathbb{Z}_2^5$ lifts

Paper 3 [Smart, 2026b] produces 32 canonical τ_σ involutions on V_{600} differing by independent $\mathbb{Z}_2$ phase choices on each of the five non-bulk right cosets, and all 32 maps swap the $K = 52$ and $K = 20$ cycle classes within each coset. We show the present trace identities are phase-independent.

Lemma 4.1 (Phase-independence). *For any τ_σ in the $\mathbb{Z}_2^5$ orbit of canonical lifts of Paper 3, the induced action on the cycle space $C_{\mathbb{Q}}$ exchanges each pair of $(K = 52, K = 20)$ cycles within a non-bulk right coset of Dic_5 . In particular the K -class projectors P_{72} and P_0 commute with the τ_σ -induced cycle action for every choice of phases, and Theorem 4.1 and Theorem 4.2 are independent of phase choice.*

Proof. Each per-coset phase swap exchanges a $K = 52$ cycle with a $K = 20$ cycle within the same right coset; it leaves the $K = 72$ and $K = 0$ bulk cycles fixed setwise (in fact pointwise). The cycle-level permutation induced by any τ_σ in the orbit therefore acts trivially on the basis vectors corresponding to $K \in \{72, 0\}$ and acts within the $K = 52 \leftrightarrow K = 20$ pair on the other ten basis vectors. In all cases the projectors P_{72} and P_0 commute with this action, so the traces of operators built from them are unchanged. $\square$

5 Numerical observation (Layer 2)

The exact rationals $13/12$ and $11/12$ from Theorem 4.1 can be compared numerically to selected published benchmarks. We tabulate published values, propagate the trace ratios, and report residuals. *No statistical inference is claimed; this section is observational only.*

We use the lensing parameter $S_8 := \sigma_8 \sqrt{\Omega_m}/0.3$ for the clustering row, since published weak-lensing tension intervals are reported as S_8 rather than bare σ_8 .

Table 1: Layer-2 numerical match. Planck 2018 baseline values from Planck Collaboration, 2020: $H_0^{\text{Planck}} = 67.36$ km/s/Mpc and $S_8^{\text{Planck}} = 0.832 \pm 0.013$.

Quantity	Trace ratio	Predicted (Layer 2)	Late-Universe value	Residual
H_0 [km/s/Mpc]	$13/12 \approx 1.0833$	72.97	73.04 ± 1.04 [Riess et al., 2022]	$\approx 0.06\sigma$
S_8	$11/12 \approx 0.9167$	0.763	$0.766^{+0.020}_{-0.014}$ [Heymans et al., 2021]	$\approx 0.24\sigma$ (KiDS)

The H_0 row uses the headline SH0ES 2022 value; multiplying the Planck value by $13/12$ gives 72.97, which is $\approx 0.06\sigma$ below the SH0ES central value at SH0ES uncertainty ± 1.04 km/s/Mpc.

The S_8 row uses the KiDS-1000 multi-probe (3×2 pt) headline $S_8 = 0.766^{+0.020}_{-0.014}$ [Heymans et al., 2021]; multiplying Planck’s $S_8 = 0.832$ by $11/12$ gives 0.763, which is below the KiDS-1000 central value. Because the predicted value lies on the *lower* side of the KiDS asymmetric interval, the directional residual uses the lower KiDS error -0.014 : $(0.766 - 0.763)/0.014 \approx 0.24\sigma$. (Including the Planck S_8 uncertainty ± 0.013 in quadrature with the lower KiDS error gives $\approx 0.18\sigma$.) The cosmic-shear-only KiDS-1000 result $S_8 = 0.759^{+0.024}_{-0.021}$ [Asgari et al., 2021] places the prediction 0.763 inside the central band on the upper side. We do NOT claim a sub- 0.1σ match for S_8 .

5.1 Ablation: rank-5 K-class corrections

For transparency we record the rank-5 alternatives: the trace ratios that would arise from $I_{C_{\mathbb{Q}}} + P_{52}$ or $I_{C_{\mathbb{Q}}} - P_{20}$ are $17/12 \approx 1.417$ and $7/12 \approx 0.583$. These rank-5 corrections give trace ratios far outside the cited H_0 and S_8 bands. Within the rank-1 pair, the trace data alone do *not* distinguish P_{72} from P_0 (both have trace 1); the assignment of the two rank-one K-classes to specific observables is the content of Hypothesis 6.1, not of the trace identity.

6 Scope and hypothesis (Layer 3)

The numerical match in Section 5 would be upgraded to a derivation only by an explicit *coupling rule* between the cycle-space operators and the cosmological observables. We state this rule as an explicit hypothesis, and we do NOT claim it is forced by the trace identity.

Hypothesis 6.1 (Cycle-mode coupling, S_8 form). *There exists a coupling map $\Gamma : \{\text{scale-rate, clustering-amplitude}\} \rightarrow \mathcal{A}_K$ such that*

$$\Gamma(\text{scale-rate}) = +P_{72}, \quad \Gamma(\text{clustering-amplitude}) = -P_0,$$

and the observed late-Universe values of H_0 and the lensing parameter S_8 relate to the early-Universe Λ CDM values by

$$H_0^{\text{late}} = \frac{\text{tr}(I_{C_{\mathbb{Q}}} + \Gamma(\text{scale-rate}))}{\text{tr}(I_{C_{\mathbb{Q}}})} H_0^{\text{early}}, \quad S_8^{\text{late}} = \frac{\text{tr}(I_{C_{\mathbb{Q}}} + \Gamma(\text{clustering-amplitude}))}{\text{tr}(I_{C_{\mathbb{Q}}})} S_8^{\text{early}}.$$

We state the hypothesis on S_8 rather than on bare σ_8 because published weak-lensing tension intervals are reported as S_8 . Translating the second relation to bare σ_8 would require an additional assumption about how Ω_m enters the coupling, which we do not make here.

We emphasise:

- Hypothesis 6.1 is *not* a theorem of this paper. It is a phenomenological postulate that, if assumed, would make the numerical match of Section 5 a parameter-free consequence of Hypothesis 6.1.
- The hypothesis is falsifiable: future H_0 measurements landing outside the 13/12-translated band, or S_8 measurements outside the 11/12-translated band, would refute Hypothesis 6.1 but leave the trace identities untouched.
- Mode-counting alone — e.g. counting orthonormal σ -antisymmetric modes on $C_{\mathbb{Q}}$ — does not by itself select the 13/12 and 11/12 ratios. Older notes in this repository (`papers/cosmological-folding` and `od1_cycle_spectrum.py`) attempted such a mode-count and reached different intermediate numbers (e.g. 17/12 from a different counting). The current paper abandons mode-counting as a route to the coupling rule and treats the rule as an explicit hypothesis. A genuine derivation would likely require an H_4 -representation decomposition of the σ -antisymmetric pair module, which is out of scope for this paper.

7 Conclusion

What is proved (Layer 1). The two operators $\hat{H} = I_{C_{\mathbb{Q}}} + P_{72}$ and $\hat{C} = I_{C_{\mathbb{Q}}} - P_0$ on the 12-dimensional rational T_{τ} -cycle space of V_{600} have exact rational traces $\text{tr}(\hat{H}) = 13$ and $\text{tr}(\hat{C}) = 11$, hence ratios 13/12 and 11/12 over the baseline $\text{tr}(I_{C_{\mathbb{Q}}}) = 12$. Within the K-saturated diagonal projector algebra, $\pm P_{72}$ and $\pm P_0$ are the only rank-one projector corrections to $I_{C_{\mathbb{Q}}}$. Both results are phase-independent across the $\mathbb{Z}_2^5$ canonical τ_{σ} orbit.

What is observed (Layer 2). Multiplying Planck 2018's $H_0 = 67.36$ by 13/12 gives 72.97 km/s/Mpc, $\approx 0.06\sigma$ below the SH0ES 2022 central value 73.04 ± 1.04 . Multiplying Planck 2018's $S_8 = 0.832 \pm 0.013$ by 11/12 gives 0.763. Since the prediction is below the KiDS-1000 multi-probe central value $0.766^{+0.020}_{-0.014}$, the directional residual uses the lower KiDS error -0.014 : $(0.766 - 0.763)/0.014 \approx 0.24\sigma$. We do not claim either match is significant beyond the level reported here.

What is hypothesised (Layer 3). Hypothesis 6.1 would upgrade Layer 2 to a derivation; this paper makes no claim to have proved it. Layers 1 and 2 stand without Layer 3.

A Verification certificate

The companion script `verify.py` performs the following exact checks using only the Python standard library and the shared `vfd_v600` package:

1. State construction: $|V_{600}| = 120$, exactly 12 T_τ -cycles each of length 10, K-multiset $\{72:1, 0:1, 52:5, 20:5\}$.
2. Projector ranks: $\text{rk } P_{72} = \text{rk } P_0 = 1$ and $\text{rk } P_{52} = \text{rk } P_{20} = 5$.
3. Resolution of identity: $P_{72} + P_0 + P_{52} + P_{20} = I_{C_{\mathbb{Q}}}$.
4. Trace identities: $\text{tr}(\hat{H}) = 13$ and $\text{tr}(\hat{C}) = 11$, both as exact `Fractions`.
5. Trace ratios: $\text{tr}(\hat{H})/12 = 13/12$ and $\text{tr}(\hat{C})/12 = 11/12$.
6. Phase-independence (Lemma 4.1): for the canonical phase-(0,0,0,0,0) τ_σ from Paper 3, AND for each of the $2^5 = 32$ phase combinations enumerated explicitly, the induced cycle action exchanges the (K=52, K=20) pairs within each non-bulk right coset and fixes the K=72 and K=0 bulk cycles; moreover the diagonal vectors of $\hat{H}$ and $\hat{C}$ are pointwise fixed under the cycle permutation (a stronger statement than trace invariance).
7. Ablation: the rank-5 alternatives $I_{C_{\mathbb{Q}}} + P_{52}$ and $I_{C_{\mathbb{Q}}} - P_{20}$ give traces 17 and 7, ratios $17/12 \approx 1.417$ and $7/12 \approx 0.583$.

Reproducibility command:

```
PYTHONPATH=papers/v600-programme/lib \
python3 papers/v600-cosmic-tensions/verify.py
```

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